

TrES-3b

R-0077

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250411 · created 2026-07-15T22:05:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460776.9302 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.022
Transit depth (Rp/Rs)^2	3.55 +/- 0.83 [%]
Orbital Inclination (inc)	81.81 +/- 0.74
Airmass coefficient 1 (a1)	1.041 +/- 0.015
Airmass coefficient 2 (a2)	-0.038 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#1 - [464, 385]
Optimal Method	PSF photometry
Transit Duration (day)	0.057 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.188 ± 0.022 vs 0.16309 (deviation 1.13σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.05σ)
Mid-time O–C	3.6 min
Depth SNR	8.5
Residual scatter	1.46 %

Light curve

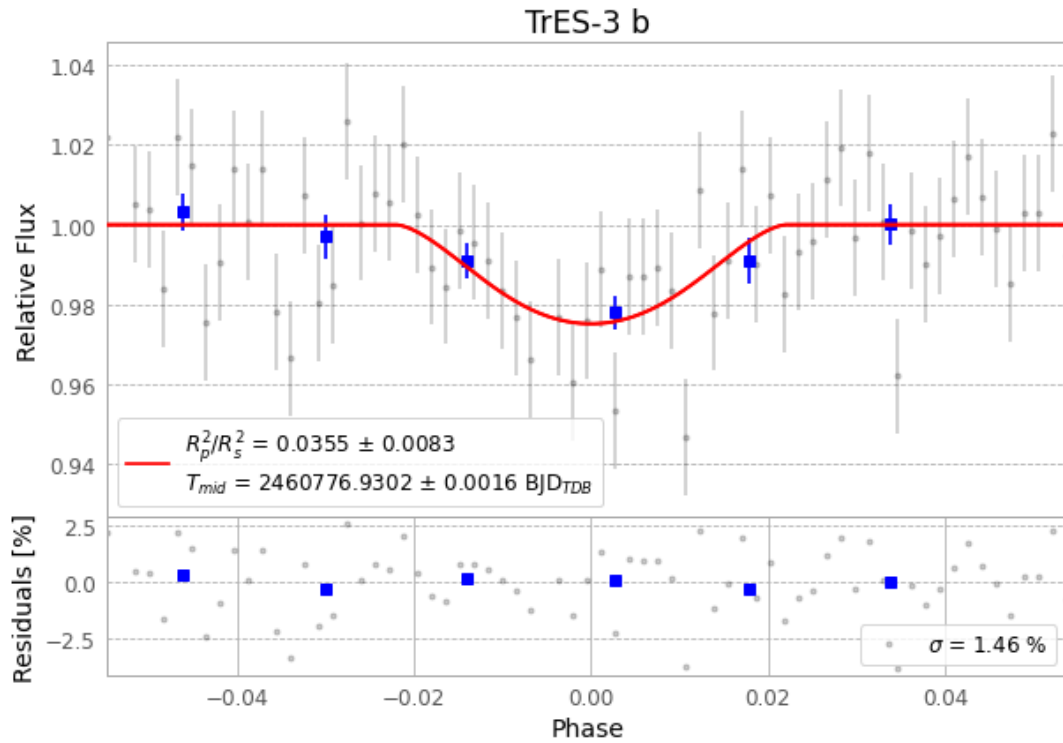


Figure 1 — FinalLightCurve_TrES-3 b_2025-04-11.png (SHA-256 61d19049fe6bf632...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d1a4bf3de9c5fa6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

69 FITS frames (45 MB), TRES-3250411083317.FITS → TRES-3250411115715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250411.log (cd2baf62bd57...), FinalLightCurve_TrES-3 b_2025-04-11.png (61d19049fe6b...), FinalLightCurve_TrES-3 b_2025-04-11.csv (d61c3ca4bcc2...)

Compute resources

Wall time 150.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:05Z.